



Path Planning for UAV Communication Networks: Related Technologies, Solutions, and Opportunities

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Path planning has been a hot and challenging field in unmanned aerial vehicles (UAV). With the increasing demand of society and the continuous progress of technologies, UAV communication networks (UAVCN) are also flourishing. The mobility of UAV nodes allows for flexible network deployment, but some challenges are brought, such as power constraints, throughput, cost, and time efficiency. Therefore, path planning is significant for UAVCN. This article presents a review of UAVCN path planning. We first introduce the network structure and performance evaluation of UAVCN. We then investigate the generic UAV path planning algorithms and the path planning algorithms in UAVCN. In this article, the advantages and disadvantages of each path planning algorithm and the functional problems. The challenges faced in path planning for UAVCN, the solutions, state-of-the-art, and representative results are also presented. In addition, we illustrate future research directions for UAVCN path planning as well, which can provide some help to researchers.

CCS Concepts: • **Computer systems organization** → *Dependable and fault-tolerant systems and networks*; • **Computing methodologies**;

Additional Key Words and Phrases: Unmanned aerial vehicle communication network, path planning, multi-UAV-assisted path planning, reinforcement learning

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1 INTRODUCTION

An **unmanned aerial vehicle (UAV)** is a non-manned aircraft controlled by combining program operating devices with radio remote control apparatus. According to different flight mechanisms, UAVs can be divided into fixed-wing, rotary-wing, and fixed / rotary-wing hybrid UAVs. Since its inception, with the breakthrough and innovation of key technologies related to the internet and

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UAV, UAVs have flourished. UAV has become an indispensable tool in modern society due to its high mobility, flexibility, simple structure, low cost, strong concealment, and high security. It will be a technology with broad commercial application prospects in the next decade. UAVs play an important role in civil fields such as express transportation [1], traffic control [2], data gathering [3], rescue tasks [4], and smart agriculture [5], widely used in military fields to perform surveillance, reconnaissance, and tracking tasks [6]. In particular, rescue tasks, disaster management [7], and data acquisition are closely related to **UAV communication networks (UAVCN)**.

In recent years, we have witnessed the popularization and use of UAVs in military and civilian fields. During this period, UAV communication networks have been extensively studied. A UAV network can be regarded as a flying wireless network. Each UAV in the network acts as a node that can send and receive messages and can as a relay for messages sent to other UAVs in the network. A UAV network can be self-organizing (without any supporting infrastructure) or supported by a ground-based or satellite-based communication infrastructure. Compared to traditional communication networks, UAV networks have some unique characteristics: (1) Nodes in UAV networks have high mobility, so network topology changes more frequently. (2) UAV networks need to ensure both UAV-to-UAV and UAV-to-infrastructure communications. (3) UAV network energy is based on UAV energy. (4) UAV networks have much more significant hardware weight constraints and space constraints.

The typical application scenarios of UAV communication include: the first is to provide ubiquitous coverage as an air **base station (BS)** [8]; the second is mobile relay technology [9]; the third is UAV-assisted information dissemination, data acquisition, surveillance [10]. Compared with the traditional communication network, UAVCN has many advantages: (1) Improve communication security. UAVs can be used as mobile relays for proximity communication to avoid information theft and improve communication security. (2) Improve communication efficiency. UAVs can be deployed flexibly. Access points equipped on UAVs can be located closer to the target user, significantly increasing transmission rates and reducing wait times. Compared with satellites, UAV communication links have less loss and relatively lower propagation delay. (3) Achieve ubiquitous communication coverage. Terrestrial communications such as mobile cellular networks have many advantages, such as high capacity and speed, but the coverage is limited. UAV can be used as an assistant of ground communication facilities to expand wireless coverage of the service area; (4) Low cost. The equipment cost and communication fee of satellite communication are relatively high. A UAV-assisted mobile communication system can cut down the network cost by reducing the requirements for fixed communication facilities. However, due to limited energy and high mobility, UAV communication is still facing challenges, such as collision avoidance, ground user requirements, energy constraints and flight times, highly dynamic network topologies, limited bandwidth, intermittent connectivity, and unpredictable noise channels. To effectively apply UAVCN to specific networks, researchers must solve many bottlenecks. For example, 3D deployment, trajectory optimization, radio and computing resource allocation, channel modelling, and path planning. One of the critical issues is path planning. From the perspective of communication, the behavior of the UAV is important. For another, the limitation of the flight time and transmit power of UAVs will seriously affect the performance of UAVCN. However, the channel state between the UAV and other communication nodes is directly related to the flight path of the UAV. In addition, the appropriate flight path may greatly shorten the communication distance. Therefore, path planning becomes an important research direction to improve communication performance and is significant to the further application of UAVs in communication. UAV path planning can be understood as obtaining a feasible flight path from the starting point to the target point based on the specific mission. Reference [11] surveyed the top 10 keywords extracted from the articles retrieved from the Web of Science and Scopus, of which the top 5 keywords were: UAV, motion

planning, UAV path planning, antenna, and path planning algorithm. The results of [11] fully reflect the importance of path planning.

Early UAV path planning work mainly focused on navigation rather than optimizing communication performance. However, the resulting trajectory may not be optimal because the communication performance is not only determined by the UAV flight trajectory but also depends on the scheduling and allocation of communication resources. Therefore, unlike traditional path planning, UAVCN path planning faces more challenges and difficulties. In addition to overcoming common path planning challenges such as path optimization, path integrity, robustness, and collision avoidance, UAVCN path planning also needs to solve communication problems, coverage, connectivity, and security privacy and reduce communication energy. Therefore, the purpose of UAVCN path planning is not only to find the best and shortest path but also to solve the performance problem, energy efficiency problem, and security and privacy protection problems of network communication. Since 2011, there has been some work on path planning for UAVs in communication scenarios. With the rapid development of artificial intelligence technology, this field is also increasingly attracting the attention of scholars. However, there is no detailed summary of UAVCN path planning so far. In such a context, we conducted this study.

The rest of this article is organized as follows: Section 2 provides the related work and our contributions. Section 3 introduces the structure and wireless technology of the UAVCN. The specific classification of general path planning algorithms is given in Section 4, and the advantages and disadvantages of each path planning algorithm are presented. Section 5 first introduces the challenges and solutions in UAVCN path planning. Besides, this section summarizes representative problems in UAVCN path planning using mentioned solutions. Next, we provide future research directions in Section 6. Finally, the conclusion of this article is given in Section 7.

2 RELATED SURVEYS AND CONTRIBUTIONS

There are very few existing surveys on UAVCN path planning. Some documents have studied the typical communication problems in the UAVCN and the common UAV path planning problems, which often study path planning separately from UAVCN. Next, we will summarize and introduce some related work.

In the existing review studies, there are many surveys and tutorials analyzing the theoretical knowledge and challenges of UAVCN. [12–21] presented some surveys on UAV communication, including the characteristics and requirements of UAV networks, main communication problems, network security, wireless charging technology, and applications. [12] provided a comprehensive survey of channel modelling in UAV communications and classified UAV channel models into deterministic, stochastic, and geometry-based models. It is pointed out that UAV communication is characterized by highly dynamic communication channels, excessive spatial and temporal variation, and airframe shadowing. In Reference [13], communication protocols, network systems, architectures, applications, and UAV positioning are investigated. In Reference [14], the authors summarized antenna design, network structure, and auxiliary communication and introduced some techniques that can enhance the performance of UAV communication. In Reference [15], the challenges of connecting UAVs with 5G/B5G cellular systems are discussed, including coverage model, ground channel, mobility, trajectory optimization, and security. The authors divided the cellular UAV network architecture into four categories: network function virtualization, mobile edge computing, the internet of things (IoT), and services. In Reference [16], three communication models (i.e., channel model, antenna model, energy consumption model) and the trajectory optimization problem are summarized. Reference [17] studied civil UAV networks from the perspective of communication and network and described the UAV network ground communication technology,

Table 1. Comparison of UAVCN Survey Papers

Ref	Year	Main features	Application	5G	Routing protocol	Network structure	Security	Path planning	Challenges
[12]	2018	Channel modelling		✓			✓		✓
[13]	2020	UAV positioning; Network structure			✓	✓	✓		✓
[14]	2020	Antenna design; Communication performance	✓	✓	✓		✓		✓
[15]	2020	3GPP advancements; Coverage; Challenges	✓	✓	✓		✓		✓
[16]	2019	Energy consumption model; Antenna model	✓	✓	✓		✓		✓
[17]	2016	Application; UAV communication	✓				✓	✓	✓
[18]	2019	Game theory; UAV networks; Interference management	✓	✓			✓	✓	
[19]	2019	Laws and regulations; Network security	✓	✓			✓		✓
[20]	2020	FANET architecture; Routing protocols				✓	✓		
[21]	2020	Reinforcement learning; Cognitive channel modeling, Path planning	✓	✓				✓	
This study	2021	A comprehensive overview UAVCNs Path planning	✓	✓		✓	✓	✓	✓

communication requirements, and application scenarios. In Reference [18], the application of game theory in UAV communication is introduced. In Reference [19], UAV cellular communication is introduced from six aspects: UAV types, key issues, UAV-assisted communication, practices, regulations, and network security. Reference [20] investigated the network communication architecture and various self-organizing routing protocols and proposed a unique network communication architecture. In Reference [21], an overview of 5G/B5G wireless networks for UAV communication is presented.

There are some review researches on path planning. In Reference [22], UAVs' various path planning techniques are classified into three categories: representative, collaborative and non-collaborative. The authors also discussed and analyzed the coverage and connectivity issues of UAVCN, but only briefly. In Reference [23], UAV path planning methods are classified as model-based, conventional, learning-based, and unit-based. The application and challenges of the UAV communication network in disaster management are also reviewed. Reference [24] mainly studies the path planning of UAVs in a **three-dimensional (3D)** environment and introduces spatial modelling problems and performance standards. Reference [25] provides an overview of research on path planning based on intelligent methods. In Reference [26], some UAV path planning algorithms based on energy-saving are described. For a quick overview, we summarize the main features and content involved in these reviews in Tables 1 and 2. Table 1 shows surveys of UAV communication networks, and Table 2 shows surveys of UAV path planning.

In Section 1, we illustrate some differences and connections between the UAVCN path planning and traditional path planning, which indicates that UAVCN path planning faces more complex environments and more significant communication performance and energy efficiency problems.

Table 2. Comparison of Path Planning Survey Papers

Ref	Year	Main features	Traditional algorithms	Intelligent algorithms	Multi-UAV path planning	Regional coverage	UAVCN
[22]	2019	Performance comparison; Comprehensive summary	✓	✓		✓	
[23]	2021	Disaster management; Algorithm statistics; 5G communication	✓	✓			✓
[24]	2019	3D UAV path planning; Workspace modelling; Path smoothing	✓	✓			
[25]	2018	Detailed analysis; Challenges		✓			
[26]	2019	Energy efficient	✓	✓			
This study	2021	Network structure; Routing protocols; Novelty classification; Detailed summary	✓	✓	✓	✓	✓

However, for UAVCN, most of the existing reviews are based on UAV cellular communication [12, 15], especially 5G and B5G, FANET [13, 20], and channel modeling [12, 16, 21]. Few reviews have studied the path planning problem in UAVCN. Although References [17, 18, 21] mentioned the path planning problem in the UAVCN, Reference [17] focuses on the requirements of UAVCN, Reference [18] focuses on the application of game theory, and Reference [19] focuses on the application of reinforcement learning. Meanwhile, there are fewer surveys on common UAV path planning, and most only review intelligent and single UAV path planning algorithms.

In contrast, traditional algorithms and some novel ideas are not summarized. We can see that there has not been a comprehensive and detailed review of UAVCN path planning to date. To the best of our knowledge, we are the first to conduct a comprehensive survey on UAV path planning and to describe the key issues in UAVCN (especially, path planning issues are discussed and analyzed detailed). Our main contributions are as follows:

- The generic path planning algorithm for UAVCN is analyzed. The generic path planning algorithm is classified into two categories: traditional algorithms and intelligent algorithms. The advantages and disadvantages of path planning algorithms are listed in tables, which will provide an important reference for researchers to choose algorithms.
- The challenges of UAVCN path planning are analyzed in detail. We are the first to comprehensively analyze the five major challenges faced by UAVCN path planning: poor security and privacy, path loss and energy limitation, substantial interference, cooperative communication and control of multiple UAVs, and complex, large-scale multi-objective optimization.
- We investigated UAVCN path planning algorithms and divided them into the heuristic algorithm, optimization theory, and reinforcement learning method. The most widely used are **block coordinate descent (BCD)** and **successive convex approximation (SCA)**, and **deep reinforcement learning (DRL)** algorithms. We also summarized the technical level and some representative results for the five main issues of energy consumption, perception of service quality, data collection and transmission, privacy and security, and coverage. The first comprehensive and detailed overview of path planning in UAVCN is provided.
- The development trend of UAV communication networks path planning is studied, which will provide a reference for future research directions in this field.

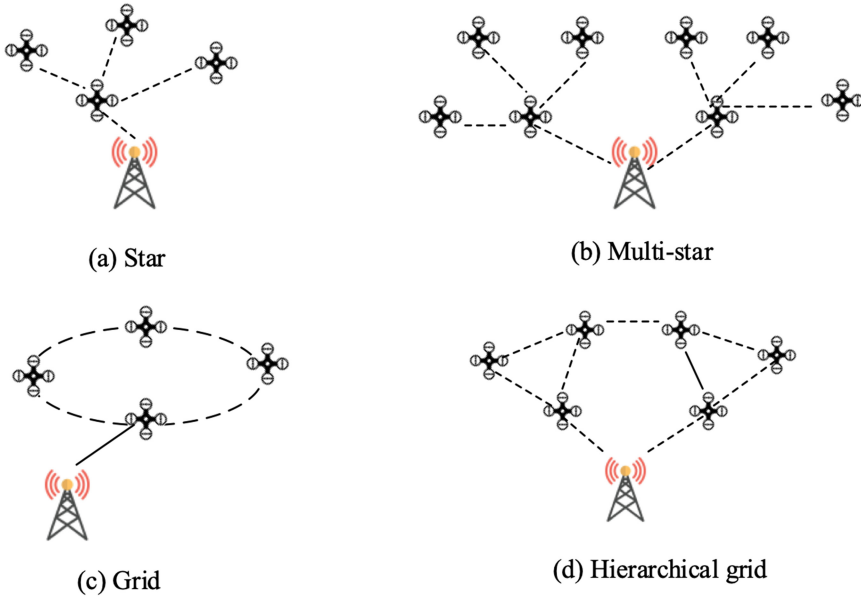


Fig. 1. Multi-UAV system.

3 BACKGROUND AND BASIS

3.1 UAV Systems and Architecture

UAV communication systems can be classified into single UAV systems and multi-UAV systems. In a single UAV system, UAV mainly communicates with ground nodes. In such networks, if a UAV fails, then it is likely to cause a communication outage. In a multi-UAV system, a variety of network topologies can be established, such as star, multi-star, grid, and hierarchical grid, as shown in Figure 1. In the star network, each UAV establishes a connection with the central node. There is no direct communication between UAV nodes but rely on the central node for relay and forwarding services. A multi-star network consists of multiple-star networks. One node in each group is connected to a ground station. A grid network removes the central node and enables all nodes to connect. Information can be directly transmitted between two nodes. A hierarchical mesh network has multiple grid networks, and one node in each group is connected to other groups. Hierarchical mesh networks support self-formation and network reorganization. When one node fails, the remaining nodes reconfigure the network. In a multi-UAV network, even if the communication link is interrupted due to the failure of some UAVs, the overall contact with the ground base station is not lost, since the network reorganizes and maintains the communication through other nodes. Therefore, the multi-UAV network is more stable and robust for UAV node faults. However, compared with a single UAV network, a multi-UAV network needs to consider more constraints, for example, the distance between UAVs and the communication range.

In addition, multilayer UAV networks have emerged [13], which can also be called UAV-assisted communication. Multi-layer UAV networks combine UAV networks with other layers such as IoT, wireless sensor networks, cloud computing, and mobile edge computing to enhance the performance of other networks.

The UAV network can rely on four communication technologies to achieve high reliability, low latency, and high throughput of UAV communication in a complex environment: (1) Direct links. Direct point-to-point communication with the ground node. If the connected UAV wants to access

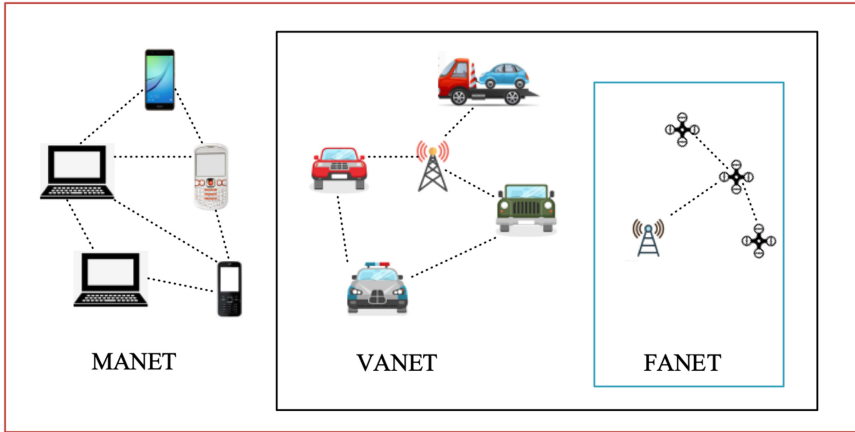


Fig. 2. MANET, VANET, and FANET.

the internet, then ground nodes need to be connected to a gateway. (2) Satellite. UAV communication by utilizing satellites is a viable option due to the global coverage of satellites. (3) Ad hoc network. A dynamically self-organizing and infrastructure-free network. (4) Cellular network. Enabling UAV communication by using cellular infrastructure and technologies. But there are few studies related to direct connection or satellites. The direct links are limited to **line-of-sight (LoS)** communication, which has a restricted operating range and environment. When using satellites for UAV communication, the long-distance between them will cause high propagation loss, delay, and high cost. Since ad hoc and cellular networks are widely used and have ubiquitous access, large-scale UAV communication can be achieved. They play important roles in UAV network communication. We will introduce UAV communication based on ad hoc and cellular networks in the following.

3.1.1 Ad Hoc Network. **Flight ad hoc network (FANET)** is a typical representative of an ad hoc network applied to the field of UAV, which can be regarded as a special form of **mobile ad hoc network (MANET)** and **vehicular ad hoc network (VANET)**. In FANET, UAVs can communicate directly without going through infrastructure. Some UAVs can be combined with manned aircraft (back-end control), ground consoles, or satellite communication. FANET has some of the same features as MANET, such as self-organizing, lack of central control, and self-organizing topology. At the same time, FANET shares VANET features such as mobility and dynamic topology changes. The main difference between both is the high node movement speed and low node density, which makes the communication design of FANET extremely difficult. Figure 2 shows their structure and connections.

FANET performance optimization has been a research hotspot in the field. The optimized self-organizing network system can give greater play to the advantages of the UAV as a network relay. For performance optimization, path planning can also be performed in addition to designing reasonable routing protocols and network topology. The main performance indicators of FANET include network coverage effectiveness, inter-node connectivity, energy efficiency, and throughput.

3.1.2 Cellular Network. Although the industry derived from UAVs is developing at high speed, new demands on the quality of communication links for UAVs are put forward. The cellular network is equipped with advanced communication technology to support GBS-UAV and U2U communication. Instead of rebuilding a new infrastructure dedicated to the UAV system, cellular-connected UAVs can reuse millions of cellular BSs globally, so it holds great promise for research

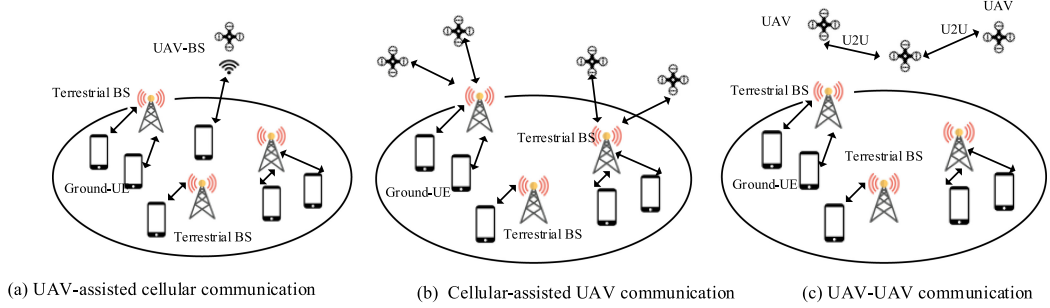


Fig. 3. UAV and cellular.

and can provide a broad commercial market. The current studies show the trend of close integration with cellular mobile communication technology. Reference [14] classifies the integration of UAVs and cellular networks: (1) UAV-assisted cellular communication. UAVs can be deployed as BS or relays to assist 5G or ground wireless communications to provide a better quality of service, higher spectral efficiency, and wireless coverage. (2) Cellular-assisted UAV communication. UAVs can be connected to cellular networks as new over-the-air user devices. Then the advanced cellular technology and the almost ubiquitous accessibility of cellular networks can be used to improve network reliability, security, and operability. (3) UAV-UAV communication. UAVs can communicate directly with each other. UAV swarms share a cellular spectrum with ground users. Figure 3 shows the integration of UAVs into cellular networks.

5G and B5G technologies, one of the advanced technology representatives of cellular networks, offer new opportunities for expanding UAV applications and services. One is that UAV plays an important role in the IoT. Second, 5G-connected UAVs play an important role in emergency communication, communication recovery, and hot spot coverage [27]. It provides satisfactory **quality of service (QoS)** and stability when ultra-high reliability and ultra-low latency communications are required. Third, UAVs can be used to improve the quality of service in 5G networks as an aid [28]. The 5G cellular mobile technology and UAVs have made it possible to realize some otherwise unimaginable ideas. However, current technology implementations do not guarantee reliable U2U communication, and standardization of 5G UAV networks remains uncertain in the coming specifications.

3.2 Path Planning

3.2.1 Execution of Path Planning. Due to the mobility of UAVs, it is possible to dynamically adjust their state to make them suitable for the communication environment by UAV path planning. To ensure the achievability, safety, and optimality of the planned path, it is necessary to use environmental information as much as possible and consider constraints such as navigation accuracy, platform maneuverability, arrival time, and fuel power limitations to avoid threats and obstacles.

UAV path planning research focuses on building mathematical models describing specific tasks, setting constraints, and optimizing methods. The following is the process of UAV path planning:

- (1) Determine the planning space according to the starting point, mission area, environmental terrain conditions, and air threats;
- (2) Determine the constraint conditions and mathematical path planning model according to the task and performance constraints.
- (3) Select or develop an appropriate search algorithm to find the optimal feasible path satisfying all constraints in the planning space;

- (4) Smooth and evaluate the searched paths;
- (5) Output result.

3.2.2 Constraints. Constraints need to be considered when solving path planning problems. Constraints related to communication design, such as throughput and delay, have been extensively studied in traditional wireless communications, while constraints on UAV movement trajectories are unique to UAV communication systems.

- **Energy constraint:** The energy of a UAV is limited. It is necessary to ensure that the UAV has completed the mission and returned safely to the intended destination before the energy is exhausted.
- **Time constraint:** Path planning must ensure that the UAV can complete the mission within a deterministic period.
- **Self-constraint:** Must meet the UAV performance constraints, including turning angle, flight speed, flight altitude, climb angle and glide angle, which are inevitable.
- **Environmental constraints:** They include flight boundaries, restricted areas (areas prohibited by regulations), and known or unknown obstacles.
- **Communication system constraints:** UAVs are limited by the limited communication range. In addition to the mobile design of the UAV communication system, the design of other parameters of the joint communication system (e.g., power control, user association, resource allocation) depends on the communication requirements.
- **Connectivity constraints:** Multi-UAV path planning needs to consider the communication connection between UAVs to avoid collisions.
- **Task constraints:** They depend on the missions performed by UAVs, such as computing offloading (throughput and quality of service), emergency communications, data collection (coverage), and the Internet of Things (energy consumption).
- **Threat:** Its size and strength will directly affect the path selection, load constraints, and weapon use of the UAV.

3.2.3 Optimization Objectives of UAV Communication System. Performance indicators are important for evaluating algorithms. The mobility of UAVs provides a broad opportunity for communication networks, but new challenges and problems also emerge. Researchers need to perform path planning for key performance indicators. In the UAVCN path planning, some key factors can be considered:

- **Time efficiency:** One of the major factors limiting the UAV path is the time spent from the initial position to the target position. Especially for UAVCN, minimizing the flight time is very important.
- **Energy consumption:** UAVs are limited by their limited battery capacity. In addition, compared with the single UAV, multi-UAV communication systems need to consume more energy, reducing the service life and communication performance of UAVs. Therefore, reducing energy consumption is especially important.
- **Coverage:** The coverage problem is to find an efficient flight path for the UAV to access all areas or vantage points within the plan area and to provide hotspot coverage.
- **Avoid Collision:** There is a risk of conflict and collision between multiple UAVs, so collisions should be avoided to ensure safety.
- **Throughput:** Throughput is the maximum data rate that a device can receive and forward. For the task of information transmission, the UAV must complete the total throughput in a certain amount of time.

- **Quality of Service:** UAVCN QoS is heavily influenced by UAV paths. The exact path of each UAV needs to be carefully planned to ensure that links with sufficient QoS are always available.
- **Interference management:** Interference management for UAVCN was first proposed by Reference [29]. Interference management needs to establish a corresponding model and an effective solution for interference events to reduce the side effects caused by interference. In multi-UAVs, interference problems are caused when UAVs reuse spectrum resources. UAVs may generate more uplink interference to adjacent cells while suffering from more downlink interference from those cells.

4 CONVENTIONAL PATH PLANNING ALGORITHMS

Even in UAV communication networks, conventional UAV path planning algorithms are still applicable, which are the basis of path planning in UAVCN. The difference between the two is that in addition to some conventional constraints, quality of service, energy efficiency, and other constraints and multiple optimization objectives must be considered in UAVCN. Therefore, it is important to classify and summarize the generic path planning algorithms.

Based on the algorithms' properties, formulas, and designs, path planning algorithms can be divided into two categories: traditional/classical and intelligent algorithms. As the name suggests, traditional methods are more classic and have been proven in other fields: An intelligent algorithm is an algorithm constructed by people imitating the laws of nature (the biological world) and has the characteristics of artificial intelligence. It can replace human brain work and has some characteristics similar to the intelligence of living beings.

4.1 Traditional Algorithm

The traditional algorithm depends on the environment information and the mathematical model. According to the properties of path planning algorithms, we divided the traditional algorithms into four categories: cell-based, mathematical model-based, node-based, and sample-based.

4.1.1 Cell-based. A cell-based algorithm means that the algorithm is self-contained. Typical examples of such algorithms are A-stars (A^*) and improved algorithms based on A^* . A^* is the algorithm to find the lowest pass-through cost. In Reference [30], the authors integrated the UAV's attitude angle information, predictive control, and real-time ideas learning into the A^* algorithm. The improved A^* realizes real-time path planning for stealth UAVs in 3D complex dynamic environments.

4.1.2 Model-based. Path planning is a constrained optimization problem that relies on mathematical models (e.g., planning models, probabilistic models). It uses mathematical equations and functions (e.g., Lyapunov functions, Bessel curves, Durbin's algorithm) for path analysis. We further classify the algorithm based on mathematical models into mathematical programming, control theory, and multi-objective optimization according to different models and functions.

The mathematical planning method first considers the dynamics constraints of the UAV; then, the optimization indicators (such as flight time, path length, energy consumption, threat level, etc.) are modeled as objective functions. Finally, the optimal value is solved by optimization calculation. Mathematical programming methods include **mixed integer programming (MLP)** [31], nonlinear programming [32], **dynamic programming (DP)** [33] and Markov decision process [34]. In some cases, DP algorithms can perform even better than **genetic algorithms (GA)**.

Path planning is essential to control the flight trajectory of the UAV. Thus, the control theory is also suitable for UAV path planning. Some control theory methods such as optimal control, model predictive control, and Lyapunov function are applied to path planning. Optimal control can reduce

the complexity of the problem, but it requires an initial solution to solve the equation and cannot be optimized. Therefore, researchers generally combine optimal control with other algorithms [35]. Model predictive control is a model used to control the UAV path planning process while satisfying a set of constraints. In Reference [36], Model predictive control is used to build a 3D UAV path planning model, and then the adaptive differential evolution method is used to plan the UAV path in real-time. Lyapunov is used to control the stability of UAVs in a dynamic environment. Lyapunov is generally used to analyze the stability of the UAV path planning algorithm [37]. Some studies also use it to reach the target trajectory [38].

4.1.3 Graph-based. A graph search control strategy finds paths on the way. We divide the graph-based path planning algorithm into deterministic search and stochastic search.

The typical representatives of deterministic search methods are depth-first search [39], breadth-first search [40], and the Dijkstra algorithm [41]. In general, the depth-first search method takes up less memory but is slower, and the breadth-first search algorithm takes up more memory but is faster. Dijkstra is a more commonly used algorithm than the former, because they are prone to NP-hard problems when dealing with more complex path planning problems. Dijkstra is derived from greedy thinking and is generally used to solve the shortest path problem. In Reference [42], the authors studied the performance comparison of Dijkstra's algorithm, Floyd's algorithm, A* algorithm, and ant colony algorithm. They demonstrated that Dijkstra's algorithm outperforms Floyd's, A* algorithm, and ant colony algorithm when running time, complexity, and path length are considered together.

Stochastic search expands the set of feasible paths in the state space by random sampling and then finds the complete feasible paths in a graph or tree structure. **Rapidly exploring random tree (RRT)** and probabilistic roadmap [43] are typical representatives of such methods. There have been many ways to improve and extend RRT, such as RRT-connect [44], RRT* [45], and hybrid RRT [46].

The deterministic method can guarantee optimality, but it is only suitable for small-scale path planning problems. When the objective function and constraint conditions are complex, it is difficult to give an effective solution. Stochastic search methods, especially the RRT algorithm, can handle UAVs' complex dynamics and kinematic constraints. However, the random sampling process of the nodes makes it difficult to guarantee the optimality of the planned route. Furthermore, stochastic methods do not apply to complex terrain or dynamic environments, because it needs to determine whether a node belongs to free space in the random sampling process.

4.1.4 Potential Field. A typical representative of the potential field method is the **artificial potential field method (APF)** [47]. APF was first produced in obstacle avoidance path planning of robots. Although the flight path obtained by the APF is not necessarily the shortest, it is the smoothest and safest. However, some shortcomings are hard to avoid, such as high computational complexity, ease of falling into local minimums, and so on. In Reference [48], dynamic APF is proposed to overcome the challenges of low performance, hardware dependence, and high computational requirements in tracking ground moving targets by multi-rotor UAVs.

For better understanding, we summarize these algorithms' advantages, disadvantages, applicable environments, and most suitable problems in Table 3.

4.2 Intelligent Algorithm

The intelligent algorithm comes from imitating some natural laws with the characteristics of artificial intelligence. It is a kind of algorithm that can replace human brain work and has some characteristics similar to the intelligence of living beings [49]. Intelligent algorithms can be divided into two heuristic forms: The first category is inspired by population change iteration and

Table 3. A Summary of Novel Traditional Path Planning Algorithms

Method	Ref	Advantage	Disadvantage	Environment
A*	[30]	Low path cost; Optimal solution	Low processing speed; Insufficient global optimization	Static
Nonlinear programming	[32]	More intuitive; Easy to implement in engineering	Difficult solution; Real-time performance	Static
DP	[33]	Simple operation; The global optimal solution	Large storage space; Prone to combinatorial explosion	Static & Dynamic
MLP	[31]	Refined route planning	A large amount of calculation; Complex modeling process	Static & Dynamic
Model predictive control	[36]	Good control effect; Strong robustness	Difficult implementation	Dynamic
Lyapunov	[35, 37, 38]	Keep stability of a UAV	Function construction	Dynamic
Depth-first search	[39]	Specific knowledge is not required; Less memory	Weak environmental adaptability; Lack of flexibility; Poor local path planning ability	Static
Breadth-first search	[40]	Specific knowledge is not required; Easy	A lot of memory; The solution obtained is not necessarily the optimal solution	Static
Dijkstra	[41, 42]	Short running time; Easy	Less efficient	Static & Dynamic
RRT	[44–46]	Probabilistic completeness; Fast calculation; Low computational cost	Random sampling process; Difficult to ensure optimality; Not suitable for complex terrain	Static
Probabilistic roadmap	[43]	Effectively avoid local minima;	Poor real-time	Static
APF	[47, 48]	Low computational effort; High real-time Smooth path	Local minima; Difficult to adjust force field; parameters	Static & Dynamic

population behavior, which can be called the swarm intelligence algorithm. The first category is based on traditional machine learning and deep learning algorithms, called artificial intelligence algorithms. Therefore, we classify intelligent algorithms into the swarm and artificial intelligence algorithms.

4.2.1 Swarm Intelligence. Various intelligent algorithms have been proposed and applied to path planning problems in recent years. How to design a fitness function considering multiple indicators, such as energy consumption, **quality of service (QoS)**, and coverage, is important. In swarm intelligence, the path planning problem is defined as exploring all feasible paths of the UAV from the starting point to the target point to minimize the total cost of the path. Some studies took some performance indicators as joint constraints, such as coverage and network data rate, and established multiple objective functions. Swarm intelligence algorithms are used to explore one or more candidate solutions for planning the path [50]. Other researchers established cost functions based on multiple indicators such as energy consumption, QoS, coverage, and so on, and then quantified these path costs. The total cost is the weighted sum of multiple sub-costs. The optimization objective is to minimize this total cost. The objective functions and constraints need to be carefully analyzed and designed according to different scenarios and tasks. The path planning parameters need to be set according to specific applications. The designer can choose from different tradeoff solutions, depending on the application requirements. It is important to note that the enhancement of one of the objective functions may lead to the deterioration of the other one. Therefore, a tradeoff choice or a reasonable set of parameters is needed according to the task requirements.

Based on the characteristics of the swarm intelligence algorithm, it can be further divided into evolutionary methods, biologically inspired methods, and meta-inspired methods according to different populations.

(1) **Evolutionary methods.** The evolutionary algorithm is based on Darwin's evolution theory and consists of selection, recombination, and mutation. In UAV path planning, the representative algorithms are GA [51], differential evolution [52], and NSGA-II [53]. Thanks to its versatility and ability to solve large-scale and nonlinear problems, GA is the most frequently used algorithm in all path planning algorithms.

The path planning problem of the fixed-wing UAV-assisted system is studied in Reference [54]. DP and convex programming methods are used to obtain the optimal path. Then the GA algorithm is used to obtain the suboptimal path to reduce the computational complexity. Since GA moves in a random walk, the diversity of the solution is reduced. Based on this, GA is improved by setting the adaptive threshold in Reference [55], and then the Bessel curve is used for path smoothing to improve the exploration ability of GA. NSGA-II is improved by using the ranking-based wheel selection and local search method in Reference [56] to avoid unnecessary saw tooth movement. This algorithm is verified by using the 3D real environment dataset of a certain area in Berlin. Unfortunately, the authors did not consider dynamic obstacles.

(2) **Biological inspiration.** Bioinspired methods mimic biological activities in nature, such as foraging. In UAV path planning, ant colony optimization, **particle swarm optimization (PSO)** [57], wolf swarm [58], grey wolf optimization, and artificial bee colony [59] are commonly used. The key to the ant colony optimization algorithm is the pheromone update state transfer and the construction of heuristic functions. Aiming at UAVs' global optimal trajectory planning, an ant colony optimization algorithm with a multi-agent structure is used for modeling and path search [60]. PSO is a way to simulate a flock of birds looking for food and is often used to plan UAV paths. Reference [61] compared the real-time path planning capability of GA and PSO for fixed-wing UAVs in a 3D dynamic environment. They conclude that genetic algorithms produce better trajectories than PSO in a statistical sense. Grey wolf optimization is a new swarm intelligent algorithm proposed by Mirjalili in 2014 [62]. This algorithm draws on the grey wolves' social organization and hunting behavior pattern and has a strong exploration ability. In Reference [63], path planning is formulated as a hard optimization problem with constraints, and grey wolf optimization is successfully applied to solve the 3D path planning problem.

Bio-inspired algorithms are computationally simple, robust, and searchable. However, as the complexity of the problem increases, these algorithms often encounter some undesirable problems, such as falling into local optimal solutions too quickly and iterating too slowly. On this basis, some algorithms improved for bio-inspired algorithms have been proposed to solve the above problems. An effective solution is to modify a mechanism and policy in the algorithm. For example, to overcome the disadvantages of slow convergence and falling into a local optimum of the original ACO, the path selection strategy and the pheromone allocation mechanism are improved in Reference [64]. In Reference [65], global search capability and local search accuracy of PSO are improved by introducing nonlinear inertia weighting coefficients. Another effective solution is combining with other path planning algorithms, which are currently widely used and suitable for improving other intelligent algorithms.

(3) **Other Meta-inspiration algorithms.** Many metaheuristic algorithms are inspired by some random phenomena in nature [49]. Bio-inspired and evolutionary methods are branches of meta-inspired algorithms, since they draw inspiration from biological activities. Tabu search [66], simulated degradation [67], clustering algorithms [68], multi-verse optimization [69] inspired by other natural phenomena. Some studies combine simulated degradation with other algorithms [69, 70]. In Reference [70], simulated degradation and PSO are combined for path planning. The results

Table 4. A Summary of Swarm Intelligence Path Planning Algorithms

Method	Ref	Advantages	Disadvantages	Environment
GA	[51, 55]	Mature development; Wide application	Poor local search ability; Difficult to determine genetic factors	Static & Dynamic
Differential evolution	[52]	Simple structure; Fast convergence; Avoids local optimal solutions	Premature convergence; Reduces diversity in later stages	Static & Dynamic
NSGA-II	[53, 56]	Multi-objective optimization; Easy to select parameters	Poor local search ability; Low population diversity; Uneven convergence distribution	Static & Dynamic
Ant colony optimization	[60, 64]	Fast convergence; Memorability	Long search time; Many parameters	Static & Dynamic
PSO	[57, 61, 65]	Few parameters; Fast convergence speed	Slow convergence speed; Low accuracy	Static & Dynamic
Grey wolf optimization	[60]	Fast convergence speed; High solution accuracy; Strong search ability	Easy to fall into local optima	Static & Dynamic
Artificial bee colony	[56]	Multi-role division of labor; No need to specify derivative	Easy to fall into local optima	Static & Dynamic
Tabu search	[63]	Step-by-step global optimization; Strong local optimization ability	Not suitable for large-scale optimization	Static & Dynamic
Simulated degradation	[64, 67, 71]	Simple calculation process; Strong versatility and robustness	Slow convergence speed; Long execution time	Static & Dynamic
Multi-verse optimization	[69, 72]	Few parameters; High search efficiency; Avoids local optima	Slow convergence; The solution accuracy is not high	Static & Dynamic
Cluster	[73]	Does not require training set Simple and fast	Susceptible to outliers	Static

show that the hybrid optimization algorithm makes PSO algorithm simpler and improves convergence speed and calculation accuracy. Since the potential path of the UAV network is similar to the universe in multi-verse optimization, it is selected to obtain a UAV path with high QoS. In Reference [72], the authors used the concepts of black holes and white holes in the universe to search and select the best path and wormholes to map interference and threats in space. Clustering algorithms are also widely used. For UAV-assisted wireless sensor networks, to minimize the cost of multiple UAV paths used for information transmission, K-means clustering is used to reduce the complexity of the problem to improve the search efficiency of subsequent path planning in Reference [73].

For better understanding, we summarize these algorithms' advantages, disadvantages, applicable environments, and most suitable problems in Table 4.

4.2.2 Artificial Intelligence. In recent years, artificial intelligence represented by reinforcement learning, artificial neural networks, game theory, and deep learning has been booming, which plays an important role in the autonomous control and decision-making of UAVs.

(1) **Reinforcement learning.** Reinforcement learning (RL) learns by interacting with the environment and then uses evaluative feedback signals to optimize decisions. RL can be divided into value function-based RL and strategy-based RL. For value function-based RL, the most commonly used algorithm is Q learning. Many studies have applied it to UAV path planning [73, 74]. However, both the state space and action space of the Q-learning algorithm are discrete. To meet the needs of the continuous state or action space, researchers proposed a **deep reinforcement**

learning algorithm (DRL) by combining deep learning and RL. DRL uses Q-network for value function approximation to solve the continuous state space problem. Unlike traditional methods that need to model UAVs, DRL directly maps the motion state of UAVs to the flight control parameters of UAVs and realizes autonomous flight planning of UAVs through repeated training.

References [76] and [77] introduced different improved DRL in UAV path planning and achieved good results. However, since the DRL action space is still discrete, the quality of the planned path can be improved. The researchers combine strategy-based RL with DL to achieve continuous state and action space. Strategy-based DRL algorithm includes a **deep deterministic policy gradient (DDPG)** [78] and distributed approximate policy optimization [79]. DDPG introduces the Actor-network based on the Q network to predict an action value that maximizes the Q value to cope with the continuous action space. In Reference [80], a phased training scheme based on DDPG is used to plan the path.

(2) **Game theory.** Game theory is the optimal strategy that can maximize the interests of all parties within the interrelated competing interests. For game theory, a game model is needed to weigh the interests of both parties to propose the optimal solution that meets the needs of both parties, and Nash equilibrium is used to reflect the stable state. It is suitable for multi-objective optimization problems with multiple constraints, especially problems with relative contradictions. It can find a mutually satisfying solution in a game between two parties, and some studies have already applied game theory to path planning. Reference [81] successfully applies game theory to the path planning of two robots. Experiments show that the game theory performs better than the popular A* algorithm. Reference [82] applies game theory to UAV path planning. A joint game-based solution is proposed where the data collected by the UAV is modeled as a hedonic joint game.

(3) **Artificial neural networks.** **Artificial neural networks (ANNs)** are algorithmic mathematical models that mimic the behavioral characteristics of animal neural networks. Compared to swarm intelligence algorithms, ANNs can solve a problem much faster. Some studies have applied ANNs to engage in UAV path planning. For example, aiming at remote UAV online path planning problem, **Hamilton-Jacobi-Bellman (HJB) neural network (NN)**-based chance control algorithm is proposed in Reference [80], which solves the HJB equations to produce suboptimal control. The BS trains the HJB-NN by downloading the current state of the UAV. Even under poor channel conditions, this method reduces the travel time and energy required by the UAV. This method is targeted for single UAV path planning, and in the future, we can consider replacing NN with a deep neural network and performing multi-UAV path planning. In Reference [84], the genetic algorithm's output is generated to train the ANN, and then the trained ANN is used for UAV obstacle avoidance path planning.

(4) **Deep learning.** Deep neural networks have excellent feature learning capability to extract concise and effective feature information from complex high-dimensional states. There are very few studies using **deep neural networks (DNN)** alone for UAV path planning, and the main reason is DL algorithms require large datasets for training. Most current research combines deep learning methods with other path planning algorithms. Some typical algorithms such as GA, PSO, and A* obtain path planning results for different scenarios to produce the dataset required for the DL algorithm. For example, to overcome the shortcomings of RRT, an algorithm called **neural RRT* (NRRT*)** is proposed in Reference [85]. The author used the trained CNN model to generate a non-uniform sampling distribution and used the RRT* method for path planning. Since CNNs require many datasets for training, the authors generated 1,070,592 images of training datasets by simulation, each of which used A* to find the best path to obtain label information.

Additionally, DL combining DL and RL can learn optimal path planning strategies by interacting with the environment. In Reference [86], a recurrent neural network extracted vital information.

Table 5. A Summary of Artificial Intelligence Path Planning Algorithms

Method	Ref	Advantages	Disadvantages	Environment
ANN	[83, 84]	Strong fault tolerance; Certain generalization and adaptive ability	Several path planning datasets; Easy to fall into local; Difficult to determine network structure	Static
Q learning	[74, 75]	Good real-time performance; Avoid local minima; Simple and efficient	Discrete state and action space; Difficult to deal with dynamic threats; Random strategy problem	Static & Dynamic
DRL	[76–80]	Model-free self-learning; Strong self-learning ability; Does not require global information; High-dimensional information perception	Not easy to converge; High sensitivity for super parameters	Static & Dynamic
Game theory	[81, 82]	Avoid conflict; Cooperation and competition; Intelligent decision	Too many Nash equilibrium solutions; Weak applicability	Static & Dynamic
DNN	[85, 86]	Large-scale parallel processing; Distributed storage; Strong adaptability and fault tolerance	A large number of datasets; High redundancy Long training time	Static & Dynamic

Then, a behavior selection strategy combining current reward value and national behavior value was developed to optimize and select paths. The algorithm fully uses local information and relative distance and can be applied to dynamic 3D UAV path planning.

For better understanding, we summarize these algorithms' advantages, disadvantages, applicable environments, and most suitable problems in Table 5.

5 UAVCN PATH PLANNING

In recent years, UAVCN path planning has gradually become a research hotspot. Compared with the early UAV path planning, UAVCN path planning faces more severe challenges because of the unique characteristics of the communication system, such as information acquisition, energy, security, data unloading, throughput, and so on. In this section, we will first elaborate on challenges in UAVCN path planning in Section 5.1. Then, in Section 5.2, we will summarize the solutions in UAVCN path planning. Finally, in Section 5.3, we will introduce the research status for the main problems using the mentioned algorithms in Section 5.2 for UAVCN path planning and sum up representative results.

5.1 Challenges

5.1.1 Poor Security and Privacy. Broadcast-based communication and line-of-sight on air-to-ground communication lead to signals easily eavesdropped on by malicious users. Additionally, some malicious UAVs may provide false information or attack other nodes to illegally obtain information and disrupt services, which may cause the entire communication system to fail. As a result, UAVCN path planning faces more serious security challenges than traditional path planning and ground communication systems. It is important to plan UAV paths to keep UAVs as far away from malicious UAVs as possible.

5.1.2 Path loss and Energy Limitation. The UAVCN consumes more energy due to implementing communication-related functions such as communication circuits and signal transmission. Furthermore, UAVs require additional propulsion power consumption. However, a UAV has limited energy due to size and weight limitations, which greatly limits its endurance and performance. It is necessary to focus on the UAV design and mission operation strategy to improve energy utilization efficiency to enable UAVs to efficiently complete tasks. On the one hand, we can install

large-capacity batteries for UAVs or install solar (wind energy) panels to improve endurance; on the other hand, the energy efficiency of UAVs can be optimized.

5.1.3 Strong Interference. The broadcast way makes UAV communication vulnerable to interference. In addition, UAVs need to reuse the spectrum of other UAVs or cellular networks because of the scarcity of spectrum resources. Therefore, UAVs may cause interference to adjacent cells on the same frequency. UAVs can also be interfered with by ground BS. However, the lack of a fixed backhaul link and control center makes it difficult to coordinate interference between cells served by UAVs. The above factors cause serious security problems and limit communication capacity. Therefore, interference management is essential to UAVCN.

5.1.4 Cooperative Communication and Control of Multiple UAVs. UAVCN often involves the coordinated communication and control of multiple UAVs. The system's overall safety and an effective mechanism should be established when planning the path. The high mobility of UAVs causes the topology of UAVCN to become dynamic and makes the UAV communication link sparse and intermittent. Therefore, it is necessary to coordinate the UAVs to ensure the reliable connection of UAVCN, and also there needs to be a link with strict delay requirements to control the UAVs in real-time to avoid collisions.

5.1.5 Complex Large-scale Multi-objective Optimization. Many factors must be considered in the path planning process to accomplish complex tasks and ensure the reliability and stability of communication. Specifically, path planning needs to meet multiple constraints, such as individual performance constraints, environment constraints, communication constraints, time constraints, collision avoidance, interference avoidance, and optimize multiple objectives, such as the smallest energy, the largest search probability, the shortest time, the largest throughput, the smallest interference, the largest security rate, and the smallest flying distance. Therefore, UAVCN path planning is a complex, large-scale, constrained, multi-objective optimization problem, which needs to be measured in many aspects.

No algorithm in the world can perfectly solve the above challenges, but we can make the best tradeoffs based on different mission requirements and communication requirements. UAVCN Path planning is a non-convex mixed-integer planning problem.

5.2 Solution

UAVCN path planning requires planning and designing UAV flight trajectories or jointly planning trajectory and communication design strategies to achieve optimal communication performance based on a given communication task. It also needs to satisfy constraints such as UAV maneuverability and communication resource limitations. Therefore, it is essentially an optimization problem with multiple constraints. The objective function corresponds to certain performance metrics to be optimized. A part of the performance metrics to be optimized can be analogous to the performance metrics of traditional terrestrial communications, such as link signal-to-noise ratio, coverage probability, throughput, time delay, and so on. UAV mission completion time and energy consumption also need to be considered.

To solve the problem of trajectory design and communication resources in UAVCN, a small part of the current work still uses common path planning algorithms. First, it is simplified to the **traveling salesman problem (TSP)**, and then some heuristic algorithms are used. But this method is only suitable for simple tasks. In UAVCN, traditional UAV path planning is often not a good solution to UAV communication problems. In such problems, the communication performance is determined by the UAV flight trajectory. It depends on the scheduling and allocation of communication resources. UAVCN path planning is a joint optimization problem of multiple variables.

There is no clear definition of path planning and trajectory optimization in UAV communication networks. Therefore, many scholars solve the UAV communication network path planning problem by joint trajectory design and optimization. The joint optimization method was proposed by Reference [87] in the scenario where UAVs are used as relays and then successfully applied to various scenarios, such as UAV-assisted communication, multiple UAV communication, and data collection scenario.

In this subsection, we will divide the solutions of UAVCN path planning into three categories: the first is heuristic algorithms [88–92]; the second is optimization theory algorithms [93–104]; the third is reinforcement learning [105–112]. Optimization theory and reinforcement learning are based on the joint optimization method.

5.2.1 Heuristic Algorithm. Heuristic algorithms use multiple iterations to search for the approximate optimal solution in the solvable space and are used primarily for UAV-assisted cellular networks and UAV-assisted sensor networks.

Reference [88] plans paths of UAVs to adapt to the random channel and achieve regional coverage. Due to D2D communication link constraints, the constructed NP-hard problem is simplified to **pick-up- and-delivery problem (PDP)** instead of TSP. Each salesperson's distance from at least one other salesperson must be less than or equal to the maximum communication. At any time, at least one salesperson needs to be on the edge of the grid. Finally, they proposed a heuristic algorithm to solve PDP. This heuristic algorithm divided the coverage area into multiple squares and assigned a UAV to each square. The divide and conquer method solves the problem of insufficient UAVs. Its disadvantage is the limitation of expandable capacity. Also, for D2D 5G network, the authors used PSO to solve the problem and used the **rank order value (ROV)** and local search to map the path planning problem to the continuous PSO particle space to solve the local minimum problem in Reference [89]. The energy transmission and data collection time allocation problem in UAVCN path planning is studied in Reference [90]. The **Karush-Kuhn-Tucker (KKT)** condition is used to solve the first sub-problem, and the DP and ant colony algorithms are combined to solve the second sub-problem.

To solve the problem that the computation of path planning increases significantly with the number of nodes in UAV-assisted wireless sensor networks, Reference [91] used the K-means algorithm to cluster the sensor nodes to reduce the problem size. The optimal detection rule is used to improve the ant colony algorithm to minimize the path cost of information transmission. For UAV networks based on backscatter communication, a path planning algorithm with a two-stage movement strategy is proposed in Reference [92]. In the first stage, the sampling-based UAV path planning algorithm is used to find the local optimal communication point. In the second stage, the median filter is used to filter out the environmental noise. The UAV jumps out of the local optimal communication point according to the RF energy field amplitude difference distribution law. The sampling-based algorithm is used to move along the symmetry axis to find the global optimal communication point.

5.2.2 Optimization Theory. The main idea of the UAVCN path planning based on optimization theory is to transform the path planning problem into several convex sub-problems and then converge to the suboptimal solution by iteration. **Block coordinate descent (BCD)** and **successive convex approximation (SCA)** are often used. But BCD and SCA have to consider the initialization value. The initial path can be solved by modeling the problem as TSP and PDP or modeling the trajectory randomly, such as an ellipsoid [93]. The access order of the ground nodes is obtained by solving the TSP or PDP, and then the optimal communication location is obtained by convex optimization. Finally, we can plan the path.

Since the UAV path is continuous, other related variables must be discretized before applying optimization theory. A widely used method is time discretization [94]: First, divide the total time into small time slots of the same length, and then optimize the position of each time slot. However, the length of time needs to be known in advance and cannot be dynamically adjusted according to different scenes for this method. Therefore, when the total length of time is not known in advance, time discretization cannot be applied. A path discretization method is used to solve this problem. Compared with time discretization, path discretization introduces time optimization variables. The flying speed of the UAV dynamically determines the length of the time slot. After obtaining the initial path and discretization, we can use SCA, BCD, and other optimization theories to obtain the optimal solution.

(1) **SCA and BCD.** BCD can divide the original non-convex problem into multiple sub-problems. In BCD, one of the sub-problems is first solved, and then the optimal solution of the sub-problem is substituted into the other sub-problems. The above process is iterated until it converges and the sub-optimal solution to the original problem can be obtained. However, when using the BCD algorithm, each sub-problem that has been decomposed may still be non-convex. SCA is a technology that transforms non-convex optimization problems into convex optimization problems, which can guarantee monotonic convergence to at least KKT. SCA generally uses the first-order Taylor expansion and introduces a slack variable to convert the original problem's non-convex objective function and constraint variables into convex and then uses the interior point method or a standard software toolkit, such as CVX, to solve the convex problem. Although the obtained solution is suboptimal, SCA can strike a balance between optimality and computational complexity. Therefore, it is the most popular algorithm for solving non-convex problems in recent years. The **poly block outer approximation (POA)** method also can solve the above problem. Still, it has been proved that the performance of the SCA method is comparable to that of the POA method, but the computational complexity of SCA is much lower [92].

In Reference [95], the 3D UAV trajectory and communication co-design to maximize throughput is studied. The joint optimization of transmission power, communication scheduling, and UAV trajectory is expressed as three concave sub-problems by using BCD and SCA: (1) use the first-order Taylor expansion to transform the transmit power optimization sub-problem into a convex optimization problem; (2) use three-layer iteration to transform the non-convex communication scheduling optimization sub-problem into a convex problem; (3) use the first-order Taylor expansion and the introduction of a relaxation factor to transform the trajectory optimization sub-problem into a convex problem; (4) use the BCD algorithm to solve these problems iteratively. Also, to maximize throughput, unlike Reference [95] applies to a single UAV system, a multi-UAV network is studied in Reference [96]. The authors investigated the joint design of launch power allocation and UAV trajectory based on SCA and BCD. By maximizing the concave lower bound of the objective function of the trajectory optimization sub-problem in the feasible convex domain until convergence, SCA is developed to solve the non-convexity of the trajectory optimization sub-problem.

In References [97] and [98], a **UAV-assisted backscatter communication network (UBCN)** is studied. Reference [97] solved the safety problem by UAV trajectories, equipment scheduling metrics, and backscatter coefficient co-design. The non-convexity of the backscattering coefficient optimization and trajectory optimization sub-problems is solved by first-order Taylor expansion and introducing relaxation variables. The issue of energy consumption is studied in Reference [98]. The UAV trajectory, uplink scheduling, and transmission power are jointly designed to maximize the energy efficiency of UBCN. The upstream scheduling optimization is a linear mixed-integer programming problem, which is solved by the cutting plane method. The transmission power optimization is transformed into an equivalent convex problem by the Dinkelbach method, and then the CVX is used to find the optimal solution. References [93] and [99] studied the security

problem and data transmission problem of the U2U communication network, respectively. In Reference [93], the fairness between users is considered. The authors maximized the minimum-security rate by user association, UAV scheduling, transmission power, and UAV trajectory co-design. The original problem is approximated as a concave function by SCA and is divided into three sub-problems by the BCD. A low-complexity relaxation method is proposed to solve the large-scale integer programming problem. Although Reference [90] investigates U2U communication with interference, the turn angle constraint of the UAV is not considered. Optimizing the trajectory of UAVs with turning angles minimizes the flight time, and relative safety is ensured in Reference [99]. Instead of using a joint optimization approach, they only consider trajectory optimization. The non-convex constraints are reformulated as convex constraints with the help of relaxation variables, Cauchy-Schwarz inequality, and Young inequality. The reformulated problems are solved iteratively by using the SCA technique.

(2) **Other methods.** Although BCD and SCA are widely used, some researchers also developed other optimization theory methods and have achieved good results. In Reference [100], information gathering and transmission are studied. Convex optimization techniques are used to maximize the QoS to study the optimal waypoints. Then, the UAV trajectory is generated by minimizing the flight time between different waypoints based on optimal control theory. Unlike using BCD to decompose non-convex problems, to solve the problem of non-convex optimization of UAV height and coverage performance, the sequence-based unconstrained minimization algorithm is used to find the global optimal UAV position and a **multi-objective evolutionary algorithm based on decomposed (MOEA/D)** is proposed to control the phase of antenna elements in Reference [101]. Then the original problem is decomposed into four sub-problems and re-formulated as a convex optimization problem with charging time as the optimization variable. The standard convex optimization method is used to solve these sub-problems.

To limit the service migration of the UAV-assisted mobile edge computing network, Reference [102] uses a directed graph to model the network topology, considers the distance, minimizes the service migration cost and the overload of the target cell and other optimization goals, and finally uses Dijkstra and prim to obtain the best path. In Reference [103], UAV trajectory and user communication scheduling co-design are studied to ensure safety. This article decomposes the original problem by introducing auxiliary variables. Then the authors developed an innovative **penalty concave-convex procedure (P-C CCP)** method and a double loop iterative algorithm to solve each sub-problem. In Reference [104], the impact of UAV jitter, user location, wind speed, and other uncertainties on UAV trajectory is fully considered. Then monotonic optimization theory and semi-definite programming relaxation are used to solve the non-convexity of the power optimization problem and obtain the best two-dimensional trajectory. But the algorithm complexity is significantly higher than SCA.

Due to space constraints, it is not possible to list all other methods. But we can learn from the above research to explore new algorithms or improve existing ones for better performance.

5.2.3 Reinforcement Learning. Reinforcement learning has been more widely used in UAVCN. Heuristic and optimization theory methods require known channel parameters and user locations, which are suitable for offline path planning. However, because the real environment is complex and changeable, likely, the model does not match, or the established mathematical model is so complex that it is difficult to solve. Not only that, it may even be impossible to establish a mathematical model that reflects the real situation due to lack of information, which makes it impossible to use convex optimization or non-convex optimization to solve the problem. RL is a promising solution that allows UAVs to interact with the environment and learn optimal policies from current and past information to make appropriate decisions.

UAVCN path planning is based on reinforcement learning. Usually, the first model of the problem is a **Markov decision process (MDP)** [105] or **stochastic game (SG)** [106]. Early, some studies used Q-learning to solve. For Q-learning, the UAV flight area is divided into a finite grid representing the finite state and action space. A Q-table is built to store the behavioral value function of each state-action pair, i.e., the Q-value. Finally, the Q-table is consulted to obtain the action policy that maximizes the Q-value. Q-learning can be effectively used for UAVCN path planning when the state and action space is minimal. Reference [107] investigates the path planning that a UAV as a BS to serve multiple users. Model-free Q-learning is used to train UAVs to make flight decisions autonomously without knowing any communication environment information. They divided the flight area into a 15×15 grid, which is used to represent the states and actions of the UAV. They also set up a Q table to store the corresponding Q value. After filling the Q form, the UAV selects actions based on the Q value. The results show that the UAV can interact with the environment to learn the trajectory. Also acting as a BS, the path planning problem is expressed as an extended multi-armed slot machine problem in Reference [108]. The authors divided the entire area into $K \times K$ equal grids, then regarded the grid as arms, the UAV as the bandit. In this method, the reward is the number of users served by the UAV on the grid, and the energy consumption is introduced to expand the multi-armed slot machine. Distance-aware upper confidence bound algorithm and ϵ -exploration is used to find the best path.

However, the system model is often complex, and state and action spaces are usually large in reality. Traditional RL is difficult to apply. For example, once the state space of Reference [107] is increased (for example, a 30×30 grid), the learning time will increase exponentially, which will lead to valuable memory and computational overhead. Therefore, DRL is usually used to solve more complex path planning problems in UAVCN. In UAVCN path planning, the state is defined as all possible states of the UAV. It is generally the position of the UAV in the 3D space, the state of the UAV itself (speed, energy), and the state of the environment (flight area of the UAV, threats, etc.); the action is defined as all possible actions of the UAV. It is generally the direction of flight of the UAV: forward, backward, left, right, up, down, hover. The power distribution ratio may also be considered in UAVCN; the reward function is defined as the interaction between the UAV and the environment. It depends on the optimization objective, i.e., the reward function is different for different purposes. For example, if the optimization objective is to avoid the obstacle zone, then UAVs will be given a fixed negative reward, a zero-value reward, and a fixed positive reward for passing through the obstacle zone, passing through the area and the target point, respectively.

Deep Q-learning (DQN) and DDPG are the most-used methods. In Reference [109], the path planning problem of UAV-assisted emergency communication network with UAVs as aerial BS to collect information is studied. The authors considered energy constraints and obstacles and converted the problem of maximizing uplink throughput into a Markov decision process. Finally, they used the DQN algorithm and Lyapunov function to plan a safe path. In Reference [109], the action is the flying direction of the UAV, the reward is the uplink throughput, and the cost is the energy consumption.

The above articles studied single UAV-assisted communication. In addition, most of the proposed methods in the existing works of heuristic algorithms and optimization theory are only suitable for single-UAV-assisted communication networks. Since the coordination and collision avoidance constraints, it is more difficult to plan the path in multi-UAV communication networks. DRL, especially the DDPG, has been successfully applied to multi-UAV communication networks. The multi-UAV path planning problem can be modeled as a game model, and then the DRL can be used to plan the path to adapt to the dynamic environment. In Reference [106], a DRL algorithm based on the **echo state network (ESN)** unit is proposed. The multi-agent path planning problem is modeled as a limited dynamic non-cooperative game model. The deep echo state network

is used to approximate the Q learning algorithm so the UAV can predict the behavior of the network. Unlike focusing or relying on coordination between UAVs, these methods are adaptation and generalization. Due to the existence of ESN, UAVs can make decisions for the invisible network state based on the rewards they received from the previous state. Unlike ESN to record the state, Reference [110] used the experience replay buffer to update and store the collected environment samples. The authors use the multi-agent consensus algorithm to solve the cooperative communication problem of multi-UAVs. The DQN algorithm was used twice to solve the problem of network connectivity and consensus convergence delay. First, it is used to deploy relay UAVs to improve the performance of the consensus algorithm, and then it is used to plan UAV paths to adapt to changes in the network and environment. More research is to use the DDPG algorithm, because it is more suitable for high-dimensional state space and more stable than DQN. In Reference [111], a UAV control strategy based on DDPG is proposed to solve the combined problem of 3D mobility and energy supply scheduling of multi-UAVs. In Reference [112], to solve the co-channel interference of multi-UAV networks and maximize throughput, a DDPG based on motion and communication co-design is proposed. The reward is the instantaneous communication rate. A coefficient punishes the UAV for flying out of the boundary and mission area, collision, and loss of connection. In the future, the distributed architecture can be adjusted to handle more UAVs based on the algorithm.

For convenience, the mentioned algorithms are summarized from a communication network, number of UAVs, optimization objectives, online/offline path planning, and 2D/3D path planning, shown in Table 6.

5.3 Technical level and representative results of main problems

In the last subsection, we have introduced various solutions to the path planning of UAVCN. To address the five challenges mentioned in Section 5.1, scholars have studied the following five main issues of UAVCN path planning: UAV secure communication [113–117], UAV Energy-efficient communication [118–129], QoS-Aware [130–134], area coverage [135–140], data collection and transmission [141–150]. These problems are the application and extension of the methods mentioned in the last subsection. Besides, we will summarize the representative results of the above-mentioned problems in UAVCN path planning.

5.3.1 UAV Secure Communication. Other UAVs can be used for interference to prevent eavesdropping and ensure safe safeguard secure communications of key UAVs. In Reference [113], two UAVs are used, one of which is used to send messages, and the other is used to send artificial noise to confuse the eavesdropper. The two UAVs can adaptively adjust their positions (i.e., trajectories) over time to facilitate secure communication and collaborative jamming. In Reference [103], one UAV is responsible for moving above the ground user and communicating with multiple users in orthogonal time division access. In contrast, the other UAV is responsible for jamming the eavesdropper. Besides, lower transmission power and limited communication range can reduce information leakage. By trajectory design, UAVs can fly close to each legitimate user to improve channel quality while moving away from eavesdroppers to reduce information leakage. Thus, confidentiality can be improved. In Reference [93], the average security rate of uplink and downlink communications can be maximized by jointly optimizing the UAV trajectory and transmitting power of UAV/ground nodes. In Reference [114], the UAV's trajectory and launch power are jointly optimized to maximize the average security rate of U2G and G2U transmissions. In addition, some works have studied the anti-interference problem [115–117]. Reference [115] first studied the path planning problem of the G2U communication system with angle constraints in the presence of potential interference. BCD and SCA are used to solve convex optimization problems. In Reference [116], the anti-jamming problem of cognitive UAVCN is solved. In the environment

Table 6. A Summary of Solutions

Ref	Communication networks	Method	Multiple/ Single UAV	Optimization objectives	On/Off-line	2D/3D
[88]	D2D + 5G	Heuristic method	Multiple	Throughput, QoS	Offline	2D
[89]	D2D + 5G	PSO, ROV, Local Search	Multiple	QoS, Cost	Offline	2D
[90]	UAV-assisted IoT	KKT, DP, AC	Single	The average age of information	Offline	3D
[91]	UAV-assisted WSN	K-means, MAXMIN ant system	Multiple	Cost	Offline	2D
[92]	UBCN	Sampling-based algorithm Median filter	Single	Communication spot	Offline	2D
[93]	D2D	BCD SCA	Single & Multiple	Secrecy rate	Offline	2D
[95]	UAV-assisted WSN	BCD, POA/ SCA	Single	Throughput	Online	3D
[96]	Multi-hop UAVs	BCD, SCA	Multiple	Throughput, Collision avoidance	Offline	2D
[97]	UBCN	BCD	Single	Uplink fair-secrecy-rate	Online	2D
[98]	UBCN	SCA, BCD, Dinkelbach, Cutting-plane technique	Single	Energy efficiency, Throughput	Offline	2D
[99]	U2U	SCA, Cauchy-Schwarz inequality, Young inequality	Multiple	Flight time	Offline	2D
[100]	UAV-Assisted IoT	Mathematical control theory, Convex optimization	Single	QoS, Fight time	Offline	3D
[101]	Wireless Power Transfer network	MOEA/D	Single	Energy harvested coverage	Offline	3D
[102]	UAV-assisted MEC	Dijkstra, Prim	Single	The number of service migrations	Offline	2D
[103]	U2U	Novel joint optimization	Single & Multiple	Secrecy rate	Offline	3D
[104]	Multiuser MISO UAV Communications	Monotonic optimization theory, Semidefinite programming relaxation	Single	Cost	Offline	2D
[107]	-	Q-learning	Single	QoS	Offline	2D
[108]	UAV-assisted Communications	Distance-aware Upper Confidence Bound, ϵ - Exploration	Single	Cost	Online	3D
[109]	UAV-assisted Communications	Safe -DQN	Single	Throughput Energy efficiency	Online	3D
[110]	Multi-UAVs	DQN	Multiple	Connectivity, Cost	Online	3D
[111]	—	UC-DDPG	Multiple	Energy Efficient, Coverage	Online	3D
[112]	Multi-UAVs	DRL	Multiple	Throughput	Online	3D

of malicious jammers with imperfect location information, the UAV trajectory and the transmission power of auxiliary transmitters are jointly optimized to maximize the information collection throughput in sensitive areas. In Reference [117], mutual interference between UAVs is also considered in addition to malicious interference. A Stackelberg game method is Besides used to solve this problem.

There are many related documents on the joint trajectory design to ensure the safety of UAVs. Many scholars have studied numerous UAV-assisted networks, such as UAV-assisted IoT, UAV-assisted cellular network, UAV-assisted sensor network, and UAV-assisted **mobile edge computing (MEC)**. We have only introduced a few, but almost all the work is carried out based on the above two aspects.

5.3.2 UAV Energy-efficient Communication. The key to improving efficiency and energy consumption is to make the path as efficient as possible. By shortening the transfer path and reducing the number of turns, we can ensure the optimal path for unmanned systems. Further, an optimal range of UAV launch power exists such that the UAV consumes the least amount of energy over the entire cycle, which can be found by a joint optimization method [118]. Existing works on energy-efficient communication of UAVs have studied communication transceiver energy consumption and UAV propulsion energy consumption (usually higher than communication power consumption).

References [119–121] studied how to reduce propulsion energy consumption. Reference [119] is the first to use a generic UAV energy model to study energy-efficient UAV communications, providing a new framework for optimizing the design of UAV trajectory parameters (e.g., instantaneous velocity and acceleration). In Reference [120], communication throughput and UAV propulsion energy consumption are considered. It shows that an energy-efficient trajectory design must balance the two. Similarly, the average power consumption of the UAV in different motion attitudes such as hovering, moving, and hovering is further studied in Reference [121]. The maximization of the service time in the service area and the minimization of the incoming/outgoing energy of the UAV are considered.

Communication transceiver power consumption mainly includes signal transmission and communication circuit consumption, which is an urgent problem in UAV-assisted IoT. IoT devices are limited to battery capacity, and UAVs are also limited by energy capacity. Many scholars have done UAV trajectory planning to solve the two problems. The mobility of UAVs can effectively reduce the energy consumption of IoT devices. In Reference [122], the problem of minimizing energy consumption of the rotary-wing UAV-assisted IoT system is studied. This article minimizes the maximum energy consumption of all equipment by co-designing communication scheduling and transmission power allocation of the equipment and the trajectory of the UAV. In Reference [123], the average data rate, total energy consumption, and coverage fairness of IoT terminals are considered to optimize the trajectory design of UAVs. The proposed on-policy and off-policy RL methods with no modules and limited actions are good at data transmission, energy efficiency, and avoiding battery drain.

In addition to UAV-assisted IoT, many scholars have studied other networks, such as UAV-assisted MEC, multi-UAV networks, and UAV-assisted sensor networks. In Reference [124], TSP and SCA with neighborhood and convex optimization technology are used to minimize the total energy consumption (including propulsion energy and communication-related energy). The problem of minimizing energy consumption of wireless power communication networks is studied in Reference [125]. The authors considered the UAV's speed, initial and final positions, energy harvesting causality, and user scheduling. For UAV-assisted mobile edge computing networks, under the constraints of the delay and energy consumption of the UAV, the trajectory and bit allocation of the UAV are jointly optimized in Reference [126]. In Reference [127], a path planning algorithm based on DRL is proposed to minimize the energy consumption of UAVs. For UAV-assisted sensor networks, to further reduce the UAV flight energy consumption, some studies have taken the communication range of the sensor nodes into account, where the UAV simply enters the communication range of the sensor nodes without flying directly over the nodes [128]. Some studies have found that finding path points at the edges of the communication range can effectively reduce the path length of the UAV, thus saving the UAV energy consumption [129].

5.3.3 Quality of Service Aware (QoS-Aware). QoS-Aware is the main problem for cellular-connected UAV networks. It can be improved from four aspects: throughput, bandwidth, delay, and data rate. UAV-assisted network path planning technology based on QoS satisfies users' unilateral

demands and considers multiple constraints, resource scheduling, QoS parameters, and other elements to maximize mission benefits. Each mission point is a network node in the information transmission process, which must achieve the corresponding QoS. The current research focuses on solving the problem of local and global QoS in the network.

In Reference [130], UAV-assisted cellular networks were studied to provide optimal QoS. A distributed algorithm based on game theory is used to solve the access point selection problem. A DRL-based algorithm is proposed to solve the UAV path planning problem that the channel state is unknown so the UAV can choose the best action based on the channel state estimation. In Reference [131], the problem of QoS-aware path planning in 5G millimeter-wave networks is studied. The approximate relationship between throughput and path loss is obtained through uniform path planning and throughput-aware UAV speed control. Finally, a QoS-aware speed control algorithm is proposed to dynamically adjust the speed of the UAV.

MEC is used to offload tasks at the network's edge and effectively reduce the burden on users and better serve them. However, user QoS is difficult to guarantee. In the UAV-assisted MEC network, Reference [132] proposes an efficient iterative algorithm to jointly optimize the resource allocation and the hovering time of UAVs in the system to minimize the energy consumption of UAVs while satisfying the QoS of IoT devices. In Reference [133], UAV power and QoS constraints are considered. Integer programming and SCA are used to optimize the user cascade scheme, UAV trajectory, and user transmission power. In real MEC networks, the number of users is often large, and the network environment is relatively complex, so DRL is more suitable for large-scale networks. In Reference [134], a DRL-based algorithm is proposed to guarantee the QoS of users in the case of ground user movement, which enables a UAV with limited energy to plan its flight trajectory according to the dynamics of the user's location to increase the total system throughput. However, UAVs can only fly at a fixed altitude, which does not make good use of the advantages of UAVs in References [133, 134]. Future research could start with this problem.

5.3.4 Area Coverage. As the name suggests, the problem of coverage path planning is to design a path that enables a UAV network, a UAV-assisted network, or a multi-UAV system to completely cover the target area. Coverage path planning is involved in many UAV applications, such as communication coverage, photogrammetry, disaster management, search and rescue, and smart agriculture. Coverage problems are mainly divided into regional coverage path planning problems and target point coverage path planning problems. The former mainly considers the complete coverage of the monitoring area and the repetition rate of the area covered by the flight path; the latter mainly ensure that the UAV can cover all target points completely without flying over the entire area. At present, the regional coverage problem is a research hotspot.

UAV coverage path planning usually solves optimization problems based on one or several optimization objectives, such as minimum energy consumption, minimum mission time, maximum coverage, and so on. Since these conditions may conflict, this problem must be solved during path planning. On the one hand, a new objective function can be created by using weight assignment. For example, in Reference [135], the maximum coverage problem with minimum mission time is considered. Due to there being a conflict between the two, a weight allocation method is used to combine these two objective functions to establish a new objective function. However, this algorithm is only applicable to static path planning. On the other hand, it can be modeled as a mixed-integer linear programming problem. For example, in Reference [136], the path planning problem with terrain constraints, obstacle avoidance, and coverage of the following flight is considered. The zigzag path method is used to generate the zigzag belt, the DE algorithm is used to optimize the path, and the fast advanced square planner generates the final path. In Reference [137], an algorithm based on the combination of distributed ANN and A* is proposed to solve the multi-UAV

coverage path planning problem. The algorithm takes into account congestion, collision avoidance, and path efficiency.

At the same time, good power and bandwidth allocation and optimal height can increase the coverage of UAVs while improving energy and resource utilization. Therefore, many studies have considered the co-design of trajectory and power. In Reference [138], satellites, UAVs, and ground communication systems are studied to enhance coverage. Under the constraints of UAV kinematics, tolerable interference, backhaul, and total energy, the UAV trajectory and in-flight transmission power are jointly optimized. Then BCD, SCA, and binary search tools are used to solve it. In Reference [139], the tradeoff between coverage and delay is discussed, and the coverage and rate analysis framework are proposed. The results show an optimal height of the UAV can best balance the total system speed and the maximum coverage probability. In addition, due to the random distribution of users, the stability of each link is very different. It is urgent to design corresponding solutions to achieve stable and maximum user coverage based on ensuring fairness. In Reference [140], the problem of jointly maximizing coverage and fairness and minimizing energy consumption is solved. The authors proposed a decentralized DRL framework to control each UAV. DNN is used for UAVs interaction and modeling, and RL is used for learning the optimal strategy. This method can cover the target area fairly in time and space and provide long-term communication coverage.

5.3.5 Data Collection and Transmission. With the civilian UAV market's rise in recent years, more researchers are applying UAVs to **wireless sensor networks (WSNs)** as data transmitters. Data collection with the assistance of UAVs can save energy and extend the usage time of the nodes in the WSNs. Unlike the general point coverage problem, data collection must consider more factors such as the communication range of sensor nodes and the data transmission rate during data collection.

The path planning problem of data collection can be decomposed into two sub-problems: the first is to find the most energy-efficient set of data collection areas, and the second is to find the optimal set of data collection points. In Reference [141], the optimal waypoints are studied by maximizing QoS based on convex optimization. Next, based on optimal control theory, the UAV trajectory is generated by minimizing the flight time between different waypoints. The main difference lies in the optimization target and the data collection method. There are many optimization objectives, such as path length [142], data collection time [143], sensor network energy consumption [144], and so on. There are also various ways of data collection. One of which is to select some cluster points and then move the nodes to these clusters to collect data [145–146]. The goal is to reduce data collection time and power consumption. However, the travel time of UAVs often exceeds the delay limit. A three-stage path planning method based on fuzzy logic is proposed to determine the maximum number of intersection points and reduce power consumption and travel time in Reference [147].

When the radio emission range of the sensor node is not considered, it can be converted to TSP. In this scenario, the mobile node must visit the exact location of each node to collect data and find the shortest path from the starting point to the endpoint. Common path planning algorithms such as genetic algorithms and ant colony algorithms can be used to solve this problem. If the data is transmitted by radio, then the mobile node does not need to visit the exact location of each node but enters its radio transmission range. This problem is similar to **TSP with neighborhoods (TSPN)**. The variable neighborhood search method is combined with the asymptotically optimal sampling PRM to solve this problem in Reference [148].

It is also necessary to consider the sensor node's data load and transmission rate. A method is proposed in Reference [149] to reduce the task completion time while considering the above factors. The path points are first initialized as the location of each sensor node, and a simulated

annealing algorithm is used to determine the traversal order of these nodes. Finally, a gradient descent method is used to reduce the data acquisition time. For multiple UAV path-planning for data collection, after selecting and clustering the best cluster heads by salp-swarm-based clustering, four UAVs are used to visit the selected CHs to collect data in different areas in Reference [150].

For convenience, the mentioned algorithms are summarized from a communication network, number of UAVs, optimization objectives, online/offline path planning, and 2D/3D path planning, which is shown in Table 7.

6 FUTURE RESEARCH DIRECTIONS

6.1 Communication Topology and Interaction Protocol

A routing protocol is one of the vital research contents of UAV communication networks, which can realize path selection in communication networks and is closely related to network performance. Unlike traditional communication networks, UAVCN has high mobility, frequent topology changes, and many interferences. Therefore, designing a more reasonable communication topology and a perfect interaction protocol is necessary. Current routing protocols for UAVCN are proposed in the context of FANET. The routing protocols of UAVCN mostly evolved from some classical protocols, such as ad hoc on-demand distance vector, destination-sequenced distance-vector, dynamic source routing, and optimized link-state routing, so the applicability of UAVCN needs to be strengthened. In the future, UAVs will have broader and more important applications. Whether multi-UAV networks or UAV-assisted communication networks require UAVs to remain fully connected and exchange information through coordination variables and functions. Some work studies the interaction between path planning and routing protocols in UAVCN. For example, Reference [75] developed UAV trajectory approximation techniques for adaptive routing in FANET networks. In Reference [72], UAV path planning is used to reduce the amount of service information about the UAV location transmitted by the FANET routing protocol. Moreover, network reconfiguration after UAV failure is an interesting area related to path planning, topological communication, and so on. Current research is focused on implementing network topology reconfigurations and is mostly focused on wireless sensor networks. No research has been found on path planning in this field. Therefore, it is necessary to study path planning and topological communication under UAV network reorganization to guarantee the destructive resistance of UAV networks.

6.2 UAV Path Planning in 3D Environments

From the spatial perspective, existing studies can be divided into 2D and 3D path planning. In the 2D scenario, it can be assumed that the UAV flies by maintaining its flight altitude or manually adjusting. The earliest path planning research was for 2D environments, because the path planning operation is simpler and less computationally intensive. In the previous decade of path planning research, 2D research accounts for a large proportion. But in recent years, due to the technology's maturity and the environment's complexity, 3D path planning has gradually become a research hotspot. Compared with 2D path planning, 3D path planning faces a more complex environment, more constraints, and kinematic constraints. Nowadays, with the development of computing chips, it can meet high-performance requirements for ML and DL. ML and DL methods can be used in 3D UAV path planning to solve NP-hard problems more accurately in a larger search space. The swarm intelligence algorithm has also made a breakthrough, which is also applicable to 3D path planning. However, the existing research is only at the initial stage, and the environment considered is still relatively simple. Therefore, there is still room for further improvements in studying 3D path planning in complex environments.

Table 7. A Summary of Representative Results of Main Problems

Ref	Communication networks	Method	Multiple/ Single UAV	Optimization targets	On/Off-line	2D/3D
[113]	–	SCA, BCD	Multiple	Secrecy rate	Offline	3D
[114]	U2G/G2U	SCA, BCD	Single	Secrecy rate	Offline	3D
[115]	–	SCA, BCD, S-procedure	Single	Robust, Throughput	Offline	2D
[116]	Cognitive UAV Communications	SCA, BCD, S-procedure	Single	Throughput	Offline	3D
[117]	–	Stackelberg game approach	Multiple	Robust	Offline	2D
[119]	Wireless communication	Sequential convex optimization, Linear state-space approximation	Single	Energy-efficiency	Offline	2D
[120]	UAV-assisted IoT	Mixed integer non-convex, Iterative algorithm	Single	Energy efficiency	Offline	2D
[121]	Free space optical communication	New feasibility check algorithm	Single	Energy-efficiency, Flight Time	Offline	2D
[122]	UAV-assisted IoT	Alternating optimization, SCA	Single	Energy consumption	Offline	2D
[123]	UAV-assisted IoT	RL	Single	Energy-efficiency	Online	2D
[124]	Wireless Communication	SCA	Single	Energy consumption	Offline	2D
[125]	Wireless Communication	Path discretization, SCA	Single	Energy consumption	Offline	2D
[126]	UAV-assisted MEC	SCA	Single	Energy consumption	Offline	2D
[127]	UAV-assisted MEC	DRL	Single	Energy consumption	Online	2D
[128]	UAV-assisted WSN	Optimal control input finding	Single	Energy-efficiency	Offline	2D
[129]	UAV-assisted WSN	Joint GA and AC	Single	Energy-efficiency	Offline	3D
[130]	UAV-assisted Cellular Networks	DRL, Game-theory-based distributed algorithm	Single	QoS, Channel path-loss	Offline	2D
[131]	5G mmWave	QoS-aware speed control	Multiple	QoS, Coverage	Offline	2D
[132]	UAV-assisted MEC	Efficient iterative algorithm	Single	QoS	Offline	2D
[133]	UAV-assisted MEC	Integer programming, SCA, Multi-variable iterative optimization	Single	Sum bits offloaded	Offline	2D
[134]	UAV-assisted MEC	Double deep Q-network	Single	QoS, throughput	Online	2D
[135]	UAV-assisted WSN	WTSC	Multiple	Coverage	Offline	2D
[136]	–	DE, Fast Marching Square	Single	Coverage	Offline	3D
[137]	Multi-UAVs	ANN, A*, K-means	Multiple	Coverage	Online	2D
[138]	Hybrid Satellite-terrestrial Networks	SCA, Bisection searching	Single	Coverage	Offline	3D
[139]	D2D	---	Single	Coverage	Offline	2D
[140]	Multi-UAV Navigation	Decentralized DRL	Multiple	Coverage	Online	2D
[141]	Multi-UAV	Convex optimization, Optimal control theory	Multiple	QoS, Flight time	Offline	3D
[147]	UAV-assisted WSN	Fuzzy Logic-based	Single	Energy consumption	Offline	2D
[148]	UAV-assisted WSN	VNS-PRM*	Single	Collected reward	Offline	2D
[149]	UAV-assisted WSN	TSP	Single	Tour time	Offline	2D
[150]	UAV-assisted WSN	DE	Multiple	Tour time	Offline	2D

However, new technologies can be used to increase the rate of information transmission. Some scholars have built big data platforms to perform dynamic path planning more efficiently. In Reference [151], Hadoop and Spark big data processing platforms are installed, where Hadoop's HDFS performs data storage, and Spark is responsible for data computation. Spark-SQL and RDD are responsible for data pre-processing.

6.3 Multi-UAV-assisted Network Path Planning

With the booming development of 5G and the Internet of Things, the scale of communication networks is increasing. For such communication scenarios with a large number of users access, multiple UAVs are needed to assist in communication. Path planning for multi-UAV-assisted communication is of great importance in future communication development. But the introduction of multiple UAVs also brings new challenges to trajectory design. In existing research, multi-UAV path planning is divided into generating and collaborative planning.

On the one hand, most studies only build the model of path generation. Therefore, how to comprehensively consider the tasks of the cluster, time coordination constraints, and track smoothing in the model to improve the accuracy of path planning is a problem worthy of further study. On the other hand, most of the current research uses swarm intelligence algorithms or hybrid algorithms such as ACO and GA. In recent years, AI methods such as DL and RL have shown good performance in sequential decision optimization problems and can realize fast path planning. Therefore, path planning of UAV clusters based on deep learning and reinforcement learning is an important research direction in the future.

Furthermore, game theory has recently been used as an effective tool for optimizing energy consumption, enhancing network coverage, resource management, coordinated control of multiple UAVs, and improving connectivity in UAVCN [152]. Game theory will provide an important foundation for distributed decision-making and cooperative behavior in UAVCN. At the same time, the inherent coupling of UAV target functions and the coexistence of multiple game scenarios due to factors such as interference or collision will also greatly facilitate the application of game theory in UAV communication networks.

6.4 Hybrid Path Planning Algorithm

Significant progress has been made in the research of UAV path planning algorithms, but each algorithm still has shortcomings. Therefore, the focus in the field remains on the exploration of new efficient path planning algorithms and fusion algorithms. For example, the combination of machine learning, swarm intelligence, and traditional algorithms in UAV path planning can effectively overcome the problem of swarm intelligence algorithms that tend to fall into local optimal solutions and improve convergence speed and robustness. The research should focus on the hybrid path planning algorithm, especially the combination of artificial intelligence and other algorithms.

6.5 AI Path Planning for Multi-source Domain Migration

The training process of deep learning and reinforcement learning path planning models is time-consuming and inefficient. Therefore, models with good generality and migration capability will greatly improve the efficiency of UAV path planning. At present, the migration effect of most artificial intelligence methods mostly depends on the degree of correlation between the source domain and the target domain. In the future, artificial intelligence methods should extract common features or parameters based on environmental samples of different source domains and characterize them in a unified state-space to achieve task migration, parameter migration, and feature migration in reinforcement learning motion planning algorithms for multiple sources. They are important guarantees for UAVs to perform tasks in more diverse environments in the future.

6.6 Security and Privacy

Due to open links and dynamic topologies, integrated networks may face malicious attacks, such as attacks, eavesdropping, hijacking, and so on. In UAV-assisted networks, security is important, because UAVs are always unattended. To avoid eavesdropping and malicious attacks, References [92, 111] use trajectory and power co-design to maximize the minimum confidentiality rate between users to ensure security. But these studies are not enough to deal with complex cyber-attacks, and cyber security is still a major challenge that must be overcome to truly utilize UAVs. Therefore, a safe and lightweight mechanism is needed to prevent and combat malicious attacks. In the latest research, generative adversarial networks are used for auxiliary optimization. The proposed generative adversarial network-based joint trajectory and power optimization with fast convergence speed can obtain nearly optimal UAV trajectory and transmission power solutions for covert communication in Reference [153]. The result proves that in addition to the optimal theory and deep reinforcement learning, other deep learning methods can also contribute to the security of the UAV network, which is very worthy of our research.

6.7 Application of High Technology

This article mentions the path planning for cellular-connected UAV networks, UAV-assisted IoT, UAV-assisted sensor networks, UAV-assisted backscatter communications, and so on. With the development of technology, more and more high technology has emerged, such as 6G, blockchain, and millimeter-wave communication. The UAV network can be protected by blockchain technology, and the 6G system will provide full support for UAVs. Besides, the UAV-assisted IoT remains an undeveloped area of future technology. However, there are few related studies, and further exploration is needed.

6.8 Improve Applicability

Many scholars have extensively studied the UAVCN path planning algorithms, but few studies have been conducted to verify the algorithms in real flight experiments. Many researchers remain at the simulation stage. However, there may be trajectory limitations, communication delays, energy constraints, electromagnetic interference, and other problems in the real environment, so the real experiments' results are different from the simulation results. Future research will tend to add more complex constraints and use more effective optimization algorithms and simulation models closer to the real thing to achieve greater feasibility and practicality of UAV path planning.

7 CONCLUSION

As a fundamental technology for UAVs to perform various tasks, path planning plays an important role in various applications of UAVs. Many related researchers have taken the characteristics of UAVCN into account based on generic path planning algorithms, thus taking the research in this field to a new level. This article provides a comprehensive survey of path planning methods for UAVCN. We have reviewed the background and history of UAV communication networks. On the one hand, we focus on analyzing generic path planning studies and compare various algorithms. Additionally, their advantages and disadvantages are concluded as well. On the other hand, we have surveyed the five challenges of path planning for UAVCN: poor security and privacy, path loss and energy limitation, strong interference, cooperative communication and control of multiple UAVs, and complex, large-scale multi-objective optimization. Then, we discussed typical solutions in UAVCN path planning by classifying the algorithms as the heuristic algorithm, optimization theory, and reinforcement learning, among which SCA, BCD, and DRL are the most popular ones.

Furthermore, based on the research in the solution in UAVCN path planning, five research directions of UAVCV are surveyed: UAV secure communication, UAV Energy Efficient communication, QoS-Aware, area coverage, and data collection and transmission, of which technical level and representative results are described, respectively. In particular, a summary of the relevant literature is presented. Finally, we look into the future research directions of UAV communication network path planning.

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